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Saveetha Institute of Medical And Technical Sciences
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Intelligent Airport Runway Safety and Aircraft Ground Operations Platform

A CAPSTONE PROJECT REPORT

Submitted partial fulfilment for the Course of

CSA1016 – Software Engineering

to receive the award of the degree

of **BACHELOR OF**

ENGINEERING IN

COMPUTER SCIENCE AND ENGINEERING

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JULY-2026

CO1 ASSESSMENT TOOL 2

Agile Sprint Planning & User Story Workshop

Project:

Aero Guard – Intelligent Airport Runway Safety and Aircraft Ground Operations Platform

Problem Statement

Airport runway and ground operations currently depend on fragmented systems and manual coordination between air traffic control, ground crews, and vehicle operators, increasing the risk of runway incursions and operational delays. Existing surveillance systems often lack real-time integration between aircraft position data, ground vehicle movement, and weather conditions, limiting proactive hazard detection. Human controllers must simultaneously monitor multiple aircraft and ground assets, which increases cognitive load and the probability of oversight during high-traffic periods.

There is minimal predictive capability to flag potential conflicts before they escalate into safety incidents, as most systems are reactive rather than preventive. Ground operation delays, such as late pushback or fueling, are not systematically tracked or correlated with runway scheduling, causing cascading flight delays. Additionally, incident and near-miss data is often recorded manually, making trend analysis and safety improvement initiatives slow and inefficient. There is a pressing need for a unified platform — AeroGuard — that consolidates runway safety monitoring, aircraft ground operations tracking, and predictive risk alerts into a single, real-time intelligent system.

1. Project Vision and Stakeholders

1.1 Project Vision

To build a unified, real-time intelligent platform that fuses aircraft and ground vehicle tracking, predictive risk analytics, and ground operations coordination into a single system — reducing runway incursions, easing controller cognitive load, and enabling airports of any size to move from reactive to preventive safety management.

1.2 Project Objectives

- Develop a real-time aircraft and ground vehicle tracking module using position and movement data across the airspace environment.
- Design a runway incursion detection system that flags potential conflicts between aircraft, vehicles, and personnel in real time.
- Build a predictive risk alert engine that identifies high-probability hazard scenarios based on traffic density, weather, and historical incident data.
- Implement a ground operations coordination dashboard to track pushback, fueling, baggage handling, and turnaround timelines.
- Create a centralized incident and near-miss logging system to replace manual reporting and support safety trend analysis.
- Enable integration of weather and visibility data to dynamically adjust safety thresholds and alert sensitivity.
- Provide role-based visualization dashboards for air traffic controllers, ground crew supervisors, and airport safety officers.
- Ensure the platform supports scalability across airports of varying size and traffic complexity.

1.3 Key Stakeholders

Stakeholder	Role / Interest
Product Owner	Owns the product vision, prioritizes the backlog, represents airport operations and regulatory interests, accepts/rejects completed work.
Scrum Master	Facilitates Scrum ceremonies, removes impediments, coaches the team on Agile practices and protects the sprint.
Development Team (Backend, Frontend, Data Engineering)	Designs, builds, and tests the tracking, detection, and dashboard modules; estimates and commits to sprint backlog items.
Air Traffic Controllers (ATC)	Primary users of the real-time tracking map and incursion alerts; rely on the system for moment-to-moment safety decisions.
Ground Crew Supervisors	Use the ground operations dashboard to coordinate pushback, fueling, baggage handling, and vehicle movement.
Airport Safety Officers	Monitor predictive risk alerts and incident/near-miss logs; drive safety audits and trend analysis.
Airport Operations Management	Business sponsor; interested in reduced delays, improved turnaround efficiency, and overall airport throughput.
Aviation Regulatory Authority	Sets safety compliance standards (e.g., incident reporting, data retention); consumes audit exports from the platform.
Ground Vehicle Operators	Field staff whose movements are tracked; rely on alerts to avoid entering active runway/taxiway zones.
UI/UX Designer	Designs role-specific, low-clutter dashboards suited to high-stress, time-critical operational environments.
QA / Test Engineers	Validate acceptance criteria for safety-critical features; perform functional, load, and failure-mode testing.
Weather Data Integration Partner	Supplies live weather/visibility feeds consumed by the predictive risk engine.

2. Epics and User Stories

The product requirements have been decomposed into seven Epics, directly mapped to the objectives in the problem statement. Each Epic is broken down into well-structured User Stories following the standard Agile format: "As a <user>, I want <feature>, so that <benefit>." Story IDs follow the pattern US-E<epic#>-<seq>.

Epic 1 (E1): Real-Time Aircraft & Ground Vehicle Tracking

ID	User Story	Priority
US-E1-01	As an air traffic controller, I want to see real-time positions of all aircraft and ground vehicles on a unified map, so that I can monitor airside movement without switching between systems.	High
US-E1-02	As an air traffic controller, I want aircraft and vehicle tracks color-coded by type and status, so that I can quickly distinguish taxiing aircraft from ground support vehicles.	High
US-E1-03	As a system administrator, I want the platform to ingest position data from multiple sensor sources (ADS-B, radar, GPS), so that tracking remains accurate and redundant.	High
US-E1-04	As an air traffic controller, I want to filter the tracking view by runway, taxiway, or zone, so that I can focus on a specific area during high-traffic periods.	Medium
US-E1-05	As a ground crew supervisor, I want to view the live location of my assigned ground vehicles, so that I can coordinate their movement efficiently.	Medium

Epic 2 (E2): Runway Incursion Detection & Conflict Alerts

ID	User Story	Priority
US-E2-01	As an air traffic controller, I want the system to automatically detect when an aircraft or vehicle enters an active runway without clearance, so that I can intervene immediately.	High
US-E2-02	As an air traffic controller, I want to receive an audible and visual alert when a potential collision path is detected between two aircraft or vehicles, so that I can prevent incidents.	High
US-E2-03	As an air traffic controller, I want incursion alerts to include distance, closing speed, and estimated time to conflict, so that I can assess severity instantly.	High
US-E2-04	As a safety officer, I want all incursion alerts logged automatically with timestamps and involved assets, so that I can review events later.	Medium

Epic 3 (E3): Predictive Risk Alert Engine

ID	User Story	Priority
US-E3-01	As an air traffic controller, I want the system to flag high-probability hazard scenarios before they occur based on traffic density and history, so that I can act preventively.	High
US-E3-02	As a safety officer, I want the predictive engine to factor in weather and visibility conditions when calculating risk scores, so that alerts reflect real operating conditions.	High
US-E3-03	As a safety officer, I want to configure risk threshold sensitivity per airport size and traffic complexity, so that alerts remain relevant and not excessive.	Medium
US-E3-04	As an air traffic controller, I want predictive alerts ranked by severity, so that I can prioritize my attention during busy periods.	Medium

Epic 4 (E4): Ground Operations Coordination Dashboard

ID	User Story	Priority
US-E4-01	As a ground crew supervisor, I want to track pushback, fueling, and baggage handling status for each aircraft in real time, so that I can coordinate turnaround activities.	High
US-E4-02	As a ground crew supervisor, I want to see estimated versus actual turnaround timelines, so that I can identify delays early.	High
US-E4-03	As a ground crew supervisor, I want to reassign ground vehicles or crew when a delay is detected, so that I can minimize cascading impact.	Medium
US-E4-04	As an airport operations manager, I want a consolidated view of all aircraft turnaround statuses, so that I can manage overall airport throughput.	Medium

Epic 5 (E5): Incident & Near-Miss Logging

ID	User Story	Priority
US-E5-01	As a safety officer, I want to log incidents and near-misses through a structured digital form, so that I can replace manual paper-based reporting.	High
US-E5-02	As a safety officer, I want near-miss events detected by the system to be pre-populated into an incident report, so that I save time on manual data entry.	High
US-E5-03	As a safety officer, I want to generate trend analysis reports from historical incident data, so that I can identify recurring risk patterns.	Medium
US-E5-04	As an airport operations manager, I want to export incident reports for regulatory audits, so that I can demonstrate compliance.	Low

Epic 6 (E6): Weather & Visibility Integration

ID	User Story	Priority
US-E6-01	As the predictive risk engine, I want to receive live weather and visibility feed updates, so that risk thresholds adjust dynamically to current conditions.	High
US-E6-02	As an air traffic controller, I want current weather and visibility conditions displayed alongside the tracking map, so that I have full situational context.	Medium
US-E6-03	As a safety officer, I want automatic alert sensitivity adjustment during low-visibility conditions, so that the system compensates for reduced human visual monitoring.	Medium

Epic 7 (E7): Role-Based Dashboards & Reporting

ID	User Story	Priority
US-E7-01	As an air traffic controller, I want a dashboard tailored to real-time tracking and incursion alerts, so that I see only information relevant to my role.	High
US-E7-02	As a ground crew supervisor, I want a dashboard focused on turnaround and ground vehicle coordination, so that I am not overwhelmed by air traffic data.	Medium
US-E7-03	As an airport safety officer, I want a dashboard summarizing incidents, near misses, and predictive risk trends, so that I can oversee overall airside safety performance.	Medium
US-E7-04	As a system administrator, I want role-based access control for each dashboard type, so that sensitive operational data is only visible to authorized users.	Medium

3. Acceptance Criteria

Acceptance criteria have been defined for the highest-priority user stories using the Given–When–Then format, ensuring each safety-critical story is specific, measurable, and testable before development begins.

US-E1-01 — Unified real-time tracking map

Given	When	Then
an air traffic controller is logged into the platform	aircraft and ground vehicles are transmitting position data	their live positions are rendered on a single map view, refreshing at least once per second
the map is displaying live tracks	the underlying data feed is lost for more than 3 seconds	the affected track is visually flagged as "stale/lost signal" rather than silently frozen

US-E1-03 — Multi-sensor data ingestion

Given	When	Then
ADS-B, radar, and GPS feeds are configured for an airport	position reports arrive from two or more sources for the same asset	the system fuses the reports into a single reconciled track with a confidence indicator
one sensor source fails	position data is still available from at least one other source	tracking continues uninterrupted using the remaining source(s) and an outage is logged

US-E2-01 — Automatic runway incursion detection

Given	When	Then
an aircraft or vehicle does not have active runway clearance	its tracked position crosses the runway boundary/hold-short line	the system raises an incursion alert to the controller within 1 second of the crossing
an incursion alert is active	the asset exits the runway boundary	the alert is automatically marked resolved and time-stamped, and the event remains in the log

US-E2-02 — Collision path alert

Given	When	Then
two tracked assets are on projected intersecting paths within the defined risk window	the system calculates a converging trajectory	an audible tone and a highlighted visual alert are triggered for the controller managing that zone
a collision-path alert is active	the controller acknowledges the alert	the alert is marked acknowledged with the controller's ID and timestamp, but remains visible until the conflict clears

US-E3-01 — Predictive hazard flagging

Given	When	Then
historical incident data, current traffic density, and asset trajectories are available	the predictive engine evaluates the current airside state	scenarios exceeding the configured risk threshold are flagged as "elevated risk" with a severity score before any actual conflict occurs

US-E4-01 — Ground operations status tracking

Given	When	Then
an aircraft has arrived at its gate	a ground crew member updates the status of pushback, fueling, or baggage handling	the dashboard reflects the updated status in real time for all users viewing that aircraft's turnaround

US-E5-01 — Structured incident logging

Given	When	Then
a safety officer needs to report an incident or near-miss	the officer completes the structured digital form (asset(s) involved, location, time, description, severity)	the report is saved to the centralized log and becomes available for trend analysis and audit export

US-E6-01 — Live weather feed integration

Given	When	Then
a weather/visibility data provider is connected	conditions change (e.g., visibility drops below a defined minimum)	the predictive risk engine automatically adjusts its alert thresholds within 60 seconds of the update

4. Product Backlog Prioritization

The product backlog is prioritized using a combination of MoSCoW categorization and a Business Value vs. Effort assessment (Value and Effort scored on a 1–10 scale, where higher value/lower effort ranks higher). Safety-critical detection and tracking features are prioritized first, since ground operations, predictive analytics, and reporting features depend on having reliable real-time positional data in place.

Rank	ID	User Story (short)	MoSCoW	Business Value (1-10)	Effort (1-10)
1	US-E1-01	Unified real-time tracking map	Must	10	6
2	US-E1-03	Multi-sensor data ingestion (ADS-B/radar/GPS)	Must	10	8
3	US-E2-01	Automatic runway incursion detection	Must	10	6
4	US-E2-02	Collision path alert (audible/visual)	Must	9	4
5	US-E3-01	Predictive hazard flagging engine (v1)	Must	9	8
6	US-E4-01	Ground ops status tracking (pushback/fuel/baggage)	Must	8	4
7	US-E5-01	Structured incident/near-miss logging	Must	8	3
8	US-E2-03	Incursion alert detail (distance/speed/ETC)	Should	8	3
9	US-E6-01	Live weather/visibility feed integration	Should	8	5
10	US-E1-02	Color-coded tracks by type/status	Should	6	2
11	US-E4-02	Estimated vs. actual turnaround timeline	Should	7	4
12	US-E7-01	ATC-focused role dashboard	Should	7	4
13	US-E2-04	Auto-logging of incursion alerts	Should	6	2
14	US-E3-02	Weather-adjusted predictive risk scoring	Should	7	5
15	US-E5-02	Auto-populate incident report from near-miss data	Should	6	4
16	US-E1-04	Filter tracking view by runway/taxiway/zone	Could	5	3
17	US-E4-03	Reassign vehicles/crew on delay detection	Could	5	4
18	US-E7-02	Ground crew supervisor dashboard	Should	6	3
19	US-E1-05	Live location of assigned ground vehicles	Could	5	3
20	US-E3-03	Configurable risk threshold sensitivity	Could	5	3

Rank	ID	User Story (short)	MoSCoW	Business Value (1-10)	Effort (1-10)
21	US-E6-02	Weather/visibility overlay on tracking map	Could	4	2
22	US-E7-03	Safety officer summary dashboard	Could	5	4
23	US-E3-04	Predictive alerts ranked by severity	Could	4	2
24	US-E6-03	Auto alert-sensitivity in low visibility	Could	4	3
25	US-E4-04	Consolidated turnaround status view	Could	4	3
26	US-E5-03	Trend analysis reports from incident data	Won't (this release)	4	5
27	US-E7-04	Role-based access control per dashboard	Should	5	3
28	US-E5-04	Export incident reports for audits	Won't (this release)	3	3

Table 4.1 — Full Product Backlog ranked by MoSCoW priority and Value/Effort. Items ranked 1–7 are selected for Sprint 1.

5. Sprint Backlog (Sprint 1)

Sprint 1 is a 3-week sprint, reflecting the higher integration and safety-validation overhead typical of aviation systems. The seven highest-priority "Must Have" stories (Rank 1–7) are selected, since together they deliver the minimum end-to-end safety loop: ingest live asset positions, display them on a unified map, detect runway incursions with alerts, produce a baseline predictive risk flag, track ground operation status, and log an incident. Each story is broken down into actionable development tasks.

US-E1-01 — Unified real-time tracking map

Task	Description	Owner	Est. (hrs)
T1.1	Design the airside tracking map UI (aircraft, vehicles, runway/taxiway overlays)	UI/UX Designer	8
T1.2	Implement real-time map rendering with WebSocket-based position updates	Frontend Dev	14
T1.3	Build backend position-broadcast service (pub/sub) for tracked assets	Backend Dev	10
T1.4	Load-test map performance with 200+ simulated concurrent tracks	QA	6

US-E1-03 — Multi-sensor data ingestion

Task	Description	Owner	Est. (hrs)
T2.1	Build ADS-B feed parser and ingestion service	Data Engineer	12
T2.2	Build radar and GPS feed adapters with common schema mapping	Data Engineer	12
T2.3	Implement track fusion/reconciliation logic with confidence scoring	Backend Dev	14
T2.4	Test failover behavior when one sensor source drops out	QA	6

US-E2-01 — Automatic runway incursion detection

Task	Description	Owner	Est. (hrs)
T3.1	Define runway/hold-short geofence boundaries per airport configuration	Backend Dev	8
T3.2	Implement clearance-status lookup and boundary-crossing detection logic	Backend Dev	12
T3.3	Build controller-facing incursion alert UI component	Frontend Dev	6
T3.4	Test detection accuracy against simulated incursion scenarios	QA	8

US-E2-02 — Collision path alert

Task	Description	Owner	Est. (hrs)
T4.1	Implement trajectory-convergence calculation (closing speed, time-to-conflict)	Backend Dev	10
T4.2	Add audible alert and highlighted visual indicator to the map UI	Frontend Dev	5

Task	Description	Owner	Est. (hrs)
T4.3	Build alert acknowledgment workflow with controller ID/timestamp	Backend Dev	5
T4.4	Test alert timing and false-positive rate under varied traffic density	QA	6

US-E3-01 — Predictive hazard flagging engine (v1)

Task	Description	Owner	Est. (hrs)
T5.1	Assemble historical incident + traffic density dataset for model training	Data Engineer	10
T5.2	Build baseline risk-scoring model (rule-based/statistical v1)	Backend Dev	14
T5.3	Expose risk flags via API and surface on controller dashboard	Frontend Dev	6
T5.4	Validate flagged scenarios against historical known-incident cases	QA	8

US-E4-01 — Ground operations status tracking

Task	Description	Owner	Est. (hrs)
T6.1	Design ground ops dashboard (pushback/fueling/baggage status per aircraft)	UI/UX Designer	5
T6.2	Implement status-update API and data model for turnaround activities	Backend Dev	8
T6.3	Build real-time status update UI for ground crew	Frontend Dev	6
T6.4	Test concurrent status updates from multiple crew members	QA	4

US-E5-01 — Structured incident & near-miss logging

Task	Description	Owner	Est. (hrs)
T7.1	Design structured incident report form (asset, location, time, severity)	UI/UX Designer	4
T7.2	Implement incident logging API and centralized data store	Backend Dev	6
T7.3	Build incident log list/search view for safety officers	Frontend Dev	5
T7.4	Test data integrity and required-field validation on submission	QA	3

6. Story Point Estimation

Story points are estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), reflecting relative complexity, effort, and uncertainty rather than absolute hours. The whole team (Product Owner facilitating, Backend/Frontend Developers, Data Engineer, QA, UI/UX) estimated independently and discussed outliers until consensus was reached.

ID	User Story	Story Points	Justification
US-E1-01	Unified real-time tracking map	8	Requires high-frequency real-time rendering of many concurrent assets with acceptable performance; moderate-high technical complexity.
US-E1-03	Multi-sensor data ingestion	13	Integrates three distinct external protocols (ADS-B, radar, GPS) with fusion/reconciliation logic; high uncertainty and dependency risk.
US-E2-01	Automatic runway incursion detection	8	Requires precise geofencing and clearance-state logic where errors have safety consequences; demands thorough edge-case handling.
US-E2-02	Collision path alert	5	Builds on existing tracking data; adds trajectory math and alert UI, moderate complexity with known patterns.
US-E3-01	Predictive hazard flagging engine (v1)	13	First version of a predictive model using historical + real-time data; high uncertainty in accuracy tuning and validation.
US-E4-01	Ground operations status tracking	5	Primarily CRUD-style status tracking with real-time sync; well-understood pattern, moderate effort.
US-E5-01	Structured incident & near-miss logging	3	Standard structured-form-to-database feature with low technical uncertainty.

Sprint 1 Total Story Points: 55

Given the safety-critical, multi-disciplinary nature of this platform, Sprint 1 is planned as a 3-week sprint with a larger cross-functional team (7–8 members including a dedicated data engineer). At an estimated velocity of 50–60 points per 3-week sprint, this scope of 55 points is considered realistic and achievable.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, the platform can ingest live aircraft and ground vehicle position data from multiple sensor sources, display it on a unified real-time tracking map, automatically detect and alert on runway incursions and collision paths, generate a baseline predictive risk flag, track ground operation turnaround status for an aircraft, and allow a safety officer to log an incident — demonstrating the platform's core real-time safety monitoring loop end-to-end."

7.2 Expected Sprint Deliverables

- A unified, real-time tracking map showing live positions of aircraft and ground vehicles, refreshing at least once per second.
- A multi-sensor ingestion pipeline (ADS-B, radar, GPS) with track fusion and source-failover handling.
- An automatic runway incursion detection module with audible/visual controller alerts and acknowledgment logging.
- A baseline (v1) predictive risk-flagging engine surfacing elevated-risk scenarios before conflicts occur.
- A ground operations dashboard tracking pushback, fueling, and baggage-handling status per aircraft.
- A structured incident and near-miss logging feature for safety officers.
- Automated unit/integration test coverage for all Sprint 1 features and a demo-ready build for Sprint Review.

7.3 Definition of Done

- Code is peer-reviewed, merged to the main branch, and passes CI build.
- Unit and integration tests are written and passing (minimum 80% coverage for new modules, with dedicated tests for safety-critical detection logic).
- Acceptance criteria for the story are verified and signed off by QA, including simulated hazard/incursion scenarios.
- UI matches approved design mock-ups and passes basic accessibility checks for high-stress operational use.
- Feature is deployed to the staging environment and demonstrated at Sprint Review.
- No critical or high-severity bugs remain open against the story, especially for detection and alerting features.

7.4 Summary

This sprint plan translates the airport runway safety and ground operations problem statement into a prioritized, estimated, and actionable Sprint 1 backlog. By focusing the first sprint on the core tracking → detection → predictive-flagging → ground-ops → incident-logging loop, the team validates the platform's central safety value proposition early, enabling faster feedback and informed planning for subsequent sprints covering advanced predictive analytics, weather integration, and role-based reporting dashboards.